



12267 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen: Ionizing Sources and ISM Conditions of $z=8-15$ Blue Monsters

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Prof. Daniel P. Stark (PI)	University of California - Berkeley
Dr. Mengtao Tang (CoI) (CoPI)	Shanghai Jiao Tong University
Dr. Joris Witstok (CoI) (ESA Member) (CoPI)	Cosmic Dawn Center, Niels Bohr Institute
Lucie Emily Rowland (CoI) (ESA Member)	Universiteit Leiden
Dr. Christina C Williams (CoI)	University of Arizona
Dr. Yoshinobu Fudamoto (CoI)	Chiba University
Dr. Hanae Inami (CoI)	Hiroshima University
Dr. Hiddo Algera (CoI)	Academia Sinica, Institute of Astronomy and Astrophysics
Prof. Ilse De Looze (CoI) (ESA Member)	Universiteit Gent
Karin Cescon (CoI) (ESA Member)	Universiteit Leiden
Dr. Kasper Elm Heintz (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Eliezer Ahrfonovich (CoI)	NASA Goddard Space Flight Center
Dr. Mengyuan Xiao (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Prof. Pratika Dayal (CoI) (CSA Member)	Canadian Institute for Theoretical Astrophysics
Rebecca Fisher (CoI) (ESA Member)	University of Manchester
Dr. Renske Smit (CoI) (ESA Member)	Liverpool John Moores University
Dr. Rychard Bouwens (CoI) (ESA Member)	Universiteit Leiden
Sander Schouws (CoI) (ESA Member)	Universiteit Leiden
Dr. Valentino Gonzalez (CoI)	Universidad de Chile

JWST Proposal 12267 (Created: Thursday, May 7, 2026, 2:00:38PM Eastern Standard Time) - Overview

Name	Institution
Prof. Charlotte Mason (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Lily Whitler (CoI) (ESA Member)	University of Cambridge, Kavli Institute for Cosmology
Dr. Zuyi Chen (CoI) (ESA Member)	University of Copenhagen, Niels Bohr Institute
Dr. Mauro Stefanon (CoI) (ESA Member)	Universitat de Valencia
Dr. Romain Alexis Meyer (CoI) (ESA Member)	University of Geneva, Department of Astronomy
Dr. Peter Senchyna (CoI)	Carnegie Institution of Washington
Dr. Rodrigo Herrera-Camus (CoI)	Universidad de Concepcion

OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
Observation Folder				
	10	uds_msa_2_v1_3	NIRSpec MultiObject Spectroscopy	(6) uds_v1
	11	uds_msa_1_v1_6	NIRSpec MultiObject Spectroscopy	(6) uds_v1
	12	par0956p2848_msa_v1_2	NIRSpec MultiObject Spectroscopy	(4) par0956p2848
	13	a2744_msa_1_v1_3	NIRSpec MultiObject Spectroscopy	(5) a2744_v1

ABSTRACT

The discovery of luminous dust-free galaxies close to cosmic dawn has been one of the biggest surprises of JWST's early science operations. The first detailed spectra suggested this population (also known as "blue monsters") may be in the midst of strong bursts, with dense and nitrogen-enriched ISM conditions that are nearly unprecedented at later epochs. However it is not clear whether these properties are representative given the small existing datasets. Fortunately, over the last year, several major JWST programs have assembled the first large spectroscopic samples of blue monsters, with redshifts ranging up to $z=14.4$. However, the prism spectra are insufficient for detailed characterization. Here we propose to obtain deep NIRSpec R=1000 rest-UV spectra of a representative sample of 10 UV-luminous galaxies between $z=8.2$ and $z=14.4$. Our galaxies are chosen to overlap with ongoing ALMA programs targeting [OIII]88um, with several of our sources selected to be bright enough for absorption line spectroscopy. This program will provide a major benchmark in our understanding of the ionizing sources and ISM conditions of the blue monsters. The spectra will definitively reveal whether dense and nitrogen-enriched ionized gas is commonplace. We will characterize the range of ionizing sources present in the UV luminous systems, testing for the presence of low metallicity stars, very massive stars, and AGN. And we will place the first constraints on the presence of outflows, which may help explain the absence of dust in the blue monsters.

OBSERVING DESCRIPTION

We propose deep NIRSpec MSA observations targeting a sample of 10 UV-luminous blue monster galaxies at $z=8.2-14.4$ (see below) using G140M and G235M gratings.

We use the MSA planning tool to design MOS configurations and plan our observations. The MSA plans are set to observe our sample of 10 blue monsters. The remaining available space on the MSAs is filled with spectroscopically-confirmed galaxies at $z>4$ identified from existing NIRSpec prism spectroscopy and shallow wide-field grism spectroscopy and imaging in our fields. We will use either the G235M/F170LP grating/filter combination or G140M/F100LP + G235M/F170LP (depending on redshift) to target the rest-frame UV emission lines (NIV] to CIII]) of the primary blue monster galaxies. Meeting our science goals requires reaching a typical 3-sigma limit on the equivalent width of emission lines in the rest-frame UV of 5Å requiring a limiting line flux of $0.3-2.1 \times 10^{-18}$ erg/s/cm², with range depending on source brightness. For gratings targeting CIII], our depth is chosen to ensure detection of both components of the doublet. For gratings targeting NIV], our depth is chosen to ensure we can detect GNz11-like NIV]/CIII]+CIV flux ratios. Our science goals require a total exposure time of 75.5 hours including overheads.

We will nod in the 3-shutter slitlet for dithering, which is appropriate for our spatially compact high redshift galaxy targets. This approach has less overhead and maximizes multiplexing efficiency, compared to the fixed or flexible dithering options that require additional MSA configurations. We set up our NIRSpec exposures using NRSIRS2, which is recommended for deep observations. We specify the position angle ranges of each MSA to (1) place all primary targets (UV-luminous blue monsters at $z > 8$) on the MSA and (2) maximize the number of additional high-priority targets, including UV-luminous high-redshift galaxies and broad-line AGN.

COSMOS-MoM-z14, $z=14.44$, $M_{UV}=-20.2$, MSA: cosmos_msa_1;
CAPERS-UDS-z11, $z=11.01$, $M_{UV}=-20.3$, MSA: uds_msa_2;
CAPERS-UDS-z10, $z=10.56$, $M_{UV}=-20.5$, MSA: uds_msa_1;
UDS-z11-2, $z=10.80$, $M_{UV}=-20.0$, MSA: uds_msa_1;
CAPERS-COSMOS-52185, $z=9.80$, $M_{UV}=-21.8$, MSA: cosmos_msa_2;
UNCOVER-3686, $z=9.32$, $M_{UV}=-21.7$, MSA: a2744_msa_1;
UNCOVER-10646, $z=8.51$, $M_{UV}=-21.6$, MSA: a2744_msa_1;
BORG-0956+2847-1130, $z=8.49$, $M_{UV}=-20.7$, MSA: par0956p2847_msa_1;
BORG-0956+2847-169, $z=8.23$, $M_{UV}=-22.3$, MSA: par0956p2847_msa_1;

JWST Proposal 12267 (Created: Thursday, May 7, 2026, 2:00:38PM Eastern Standard Time) - Overview

BORG-0956+2847-169-neigh, $z=8.21$, $M_{UV}=-21.5$, MSA: par0956p2847_msa_1.

Proposal 12267 - Targets - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen: Ionizin...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(4)	par0956p2848	RA: 09 56 27.2424 (149.1135100d) Dec: +28 48 44.06 (28.81224d) Equinox: J2000		
	Comments: Description=[]				
	(5)	a2744_v1	RA: 00 14 22.2910 (3.5928792d) Dec: -30 24 9.94 (-30.40276d) Equinox: J2000		
	Comments: Description=[]				
	(6)	uds_v1	RA: 02 17 28.4802 (34.3686675d) Dec: -05 08 52.84 (-5.14801d) Equinox: J2000		
	Comments: Description=[]				

Proposal 12267 - Observation 10 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen:...

Observation	Proposal 12267, Observation 10: uds_msa_2_v1_3										Thu May 07 19:00:38 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: We select specific position angles to (1) place all primary targets (UV-luminous blue monsters at z > 8) on the MSA and (2) maximize the number of additional high-priority targets, including UV-luminous high-redshift galaxies and broad-line AGN.										
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	uds_v1	RA: 02 17 28.4802 (34.3686675d) Dec: -05 08 52.84 (-5.14801d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	uds_v1 (2036 sources)		jwst-nirspec-g235m		1.5	
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	34.428101541666 66 Degrees - 5.128170833333 43 Degrees	39.994693926286 125			3	12	26435.069

Proposal 12267 - Observation 10 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen:...

Special Requirements	Aperture PA Range 35 to 45 Degrees (V3 256.4254303 to 266.4254303) MSA Planned Aperture PA 40.0000 to 40.0000 Degrees (V3 261.4254303 to 261.4254303)
----------------------	--

Proposal 12267 - Observation 11 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen:...

Observation	Proposal 12267, Observation 11: uds_msa_1_v1_6										Thu May 07 19:00:38 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: We select specific position angles to (1) place all primary targets (UV-luminous blue monsters at $z > 8$) on the MSA and (2) maximize the number of additional high-priority targets, including UV-luminous high-redshift galaxies and broad-line AGN.										
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	uds_v1	RA: 02 17 28.4802 (34.3686675d) Dec: -05 08 52.84 (-5.14801d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	uds_v1 (2036 sources)		jwst-nirspec-g235m		1.5	
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	34.265974541666 665 Degrees - 5.1212466666666 5 Degrees	195.00907532736 048			3	12	25559.735

Proposal 12267 - Observation 11 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen:...

Special Requirements	Aperture PA Range 190 to 197 Degrees (V3 51.4254303 to 58.4254303) MSA Planned Aperture PA 195.0000 to 195.0000 Degrees (V3 56.4254303 to 56.4254303)
----------------------	--

Proposal 12267 - Observation 12 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen:...

Observation	Proposal 12267, Observation 12: par0956p2848_msa_v1_2										Thu May 07 19:00:39 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: We select specific position angles to (1) place all primary targets (UV-luminous blue monsters at z > 8) on the MSA and (2) maximize the number of additional high-priority targets, including UV-luminous high-redshift galaxies and broad-line AGN.										
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	par0956p2848	RA: 09 56 27.2424 (149.1135100d) Dec: +28 48 44.06 (28.81224d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	par0956p2848 (4 sources)		jwst-nirspec-g140m	1.5			
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	149.09880979166 667 Degrees 28.792863333333 337 Degrees	59.992871148615 16			3	6	7527.867

Special Requirements	Aperture PA Range 55 to 65 Degrees (V3 276.4254303 to 286.4254303) MSA Planned Aperture PA 60.0000 to 60.0000 Degrees (V3 281.4254303 to 281.4254303)
----------------------	--

Proposal 12267 - Observation 13 - BLUEMOON - Benchmarking Luminous UV-Emitting Monsters Observed from Oxygen to Nitrogen:...

Observation	Proposal 12267, Observation 13: a2744_msa_1_v1_3										Thu May 07 19:00:39 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: We select specific position angles to (1) place all primary targets (UV-luminous blue monsters at z > 8) on the MSA and (2) maximize the number of additional high-priority targets, including UV-luminous high-redshift galaxies and broad-line AGN.										
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	a2744_v1	RA: 00 14 22.2910 (3.5928792d) Dec: -30 24 9.94 (-30.40276d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	a2744_v1 (4416 sources)				jwst-nirspec-mr	1.5
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F100LP)	c1	3 Shutter Slitlet	3.608872625 Degrees - 30.411191944444 454 Degrees	34.991882500407 286			3	6	21095.535
	2	2 (G235M/F170LP)	c1	3 Shutter Slitlet	3.608872625 Degrees - 30.411191944444 454 Degrees	34.991882500407 286			3	6	3588.867

Special Requirements	Aperture PA Range 30 to 40 Degrees (V3 251.42543030000002 to 261.4254303) MSA Planned Aperture PA 35.0000 to 35.0000 Degrees (V3 256.4254303 to 256.4254303)
----------------------	---